

EXPERIMENTAL QUANTUM MECHANICS RESEARCH PROJECT

Experimental Investigation of Quantum Polarization and Malus' Law

A practical research pathway toward a documented quantum-physics portfolio, mentorship, and scholarship applications

Prepared as a student research proposal and portfolio plan

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Project at a Glance

Item	Plan
Project type	Experimental quantum optics / quantum mechanics
Core measurement	Polarization-dependent transmitted intensity
Main law tested	Malus' law: $I(\theta) = I_0 \cos^2\theta$
Primary output	Experimental dataset + uncertainty analysis + research report
Advanced extension	Low-light/single-photon polarization measurement
Long-term extension	Photon polarization correlations / Bell-inequality experiment under expert supervision
Main software	Python, NumPy, pandas, Matplotlib, SciPy
Target use	Research portfolio, mentorship, summer research and scholarship applications

1. Executive Summary

This project proposes a real experimental investigation of the polarization of light and the quantitative relationship described by Malus' law. The central goal is to determine how closely measured transmitted intensity follows the quantum-mechanical prediction $I(\theta) = I_0 \cos^2\theta$, while identifying and quantifying experimental uncertainties and systematic effects.

The project is deliberately designed in two stages. Stage A can be performed with accessible optical equipment and establishes rigorous experimental practice. Stage B, if a university or research laboratory becomes available, upgrades the experiment to low-light or single-photon measurements. This makes the project both realistic to start and capable of becoming an advanced quantum-optics research experience.

The final portfolio should contain a research proposal, laboratory notebook, raw data, calibrated measurements, Python analysis, uncertainty budget, final report, poster, presentation, and a short research summary for scholarship or research applications.

2. Research Question

Primary question: To what accuracy does experimentally measured transmitted optical intensity through two linear polarizers follow the quantum-mechanical polarization prediction $I(\theta) = I_0 \cos^2\theta$, and which experimental effects dominate the deviation from the ideal model?

Extension question: When the experiment is performed at sufficiently low light levels, can the same polarization probability law be tested using photon-counting measurements rather than ordinary power/intensity measurements?

3. Hypothesis

H1: The measured transmitted intensity will follow a $\cos^2$ dependence on the relative angle between the polarization axes, within the combined experimental uncertainty.

H2: After calibration and correction for background signal, the dominant deviations from the ideal model will be attributable to imperfect polarizers, detector response, alignment, ambient/background light, finite angular resolution, and source stability.

H3: In a properly supervised low-light extension, photon-counting statistics will reproduce the same polarization probability dependence, with additional statistical fluctuations consistent with counting statistics.

4. Scientific Background

Polarization describes a property of electromagnetic radiation associated with the orientation of its electric-field oscillation. A linear polarizer transmits one polarization component and attenuates the orthogonal component.

For linearly polarized incident light and an ideal analyzer at relative angle θ , the transmitted intensity is given by Malus' law: $I(\theta) = I_0 \cos^2\theta$. In a quantum description, polarization is a two-state degree of freedom; measurement probabilities are determined by the overlap between the prepared state and the measurement basis. The probability

corresponding to transmission through an analyzer has the same $\cos^2$ form.

This project therefore connects a measurable laboratory quantity to a central quantum-mechanical probability rule. The first stage does not by itself prove every aspect of quantum mechanics; it tests a quantitative prediction associated with polarization and prepares the student for genuinely quantum optical measurements.

5. Why This Is a Strong Student Research Project

The project is stronger than a simple demonstration because it asks a quantitative research question, requires calibration, produces a substantial dataset, compares a theoretical model with measurements, and explicitly studies uncertainty and systematic error.

The project also develops transferable research skills: experimental design, safe laboratory practice, scientific writing, Python programming, curve fitting, uncertainty propagation, data visualization, reproducibility, and scientific communication.

A high-quality student project should not claim that it has discovered new physics unless the evidence genuinely supports such a claim. The strongest outcome is a careful, reproducible measurement and an honest analysis of its limitations.

6. Experimental Design — Stage A

Conceptual setup: light source → polarizer (P1) → rotatable analyzer (P2) → detector → data-acquisition system → computer/Python analysis.

Recommended measurement range: rotate P2 through 0° – 180° in fixed angular steps, for example 5° or 10° . At each angle, record detector signal for a fixed integration time. Repeat selected points to estimate repeatability.

Before taking the main data, measure detector dark/background signal, verify source stability, check polarizer alignment, and record the actual angle rather than assuming the mechanical scale is perfect.

The project should use a low-power, laboratory-appropriate light source and suitable optical components. A laser should never be viewed directly or used without appropriate laser-safety procedures. If a supervised laboratory has safer alternatives, follow the laboratory's rules.

7. Equipment — Stage A

Core equipment: stable light source; two linear polarizers; rotation mount or angular scale; optical mounts; detector or light sensor; breadboard/optical rail or stable table; ruler/alignment aids; computer or phone-connected data system where appropriate; notebook.

Preferred additions: calibrated photodiode/power sensor, neutral-density filters, iris, beam stop, dark enclosure, and a digital angle readout.

Low-cost route: a suitable LED/laser source, inexpensive polarizing filters, a photodiode/light sensor module, and a microcontroller can be used for an educational prototype. The final scientific measurements should be made with calibrated equipment whenever possible.

8. Experimental Procedure

1. Mount the source, P1, P2, and detector so the beam is stable and approximately centered.
2. Establish a safe and repeatable alignment. Record the source, detector, polarizers, distances, and environmental conditions.
3. Measure the detector background with the source blocked or otherwise following the detector manufacturer's recommended dark procedure.
4. Fix P1 and define its transmission axis as 0° . Determine the analyzer reference angle carefully.
5. For each chosen angle θ , record detector output for the same integration time. Take repeated readings at selected angles and repeat the full scan if possible.

6. Subtract the measured background according to the detector model. Do not remove data points simply because they do not match the theory.
7. Normalize the corrected intensity when appropriate and fit the data to $A \cos^2(\theta - \theta_0) + B$, where A , θ_0 , and B are fitted parameters.
8. Compare the fitted model with the ideal $\cos^2$ prediction and report residuals and uncertainty.

9. Data Table Template

Record at minimum: trial number; analyzer angle θ ; raw detector reading; background reading; corrected signal; integration time; repeat number; environmental notes; estimated uncertainty.

A useful spreadsheet/Python structure is: angle_deg, raw_signal, background, corrected_signal, uncertainty, repeat_id.

10. Mathematical Analysis

Ideal model: $I(\theta) = I_0 \cos^2 \theta$.

Practical fit model: $I_{\text{fit}}(\theta) = A \cos^2(\theta - \theta_0) + B$, where A represents the scale, θ_0 represents angular offset, and B represents a residual baseline/background term.

Residual: $r_i = I_i - I_{\text{fit}}(\theta_i)$. Plot residuals against angle to identify systematic structure.

Goodness of fit: calculate a reduced chi-square when reliable uncertainties are available: $\chi^2_{\text{red}} = [1/(N-p)] \sum [(I_i - I_{\text{fit},i})/\sigma_i]^2$. Here N is the number of measurements and p is the number of fitted parameters.

For repeated measurements, estimate the standard deviation and standard error where appropriate. Propagate uncertainty from detector noise, angle uncertainty, repeatability, and calibration uncertainty rather than reporting only a percentage difference.

11. Python Analysis Plan

Use Python with NumPy, pandas, Matplotlib, and SciPy where available.

The analysis script should: import raw data; calculate background-corrected signal; convert degrees to radians; fit the $\cos^2$ model; calculate parameter uncertainties where supported; plot raw and fitted data; plot residuals; calculate summary statistics; and export tables/figures.

Keep the raw dataset unchanged. Store the analysis code separately and record the software version and date so the result is reproducible.

12. Uncertainty and Systematic-Error Plan

Potential random effects: detector noise, short-term source fluctuations, reading variability, and environmental fluctuations.

Potential systematic effects: imperfect polarizer extinction ratio, polarizer-axis offset, detector nonlinearity, detector spectral response, beam misalignment, stray light, source polarization, mechanical backlash, and incorrect background subtraction.

A strong report should rank the dominant uncertainty sources and explain which ones were measured, which were estimated, and which could be reduced in a future experiment.

13. Stage B — Advanced Quantum-Optics Extension

If a university laboratory and qualified supervisor are available, upgrade the experiment to low-light or single-photon measurements using appropriate photon-counting equipment. The purpose is to test the polarization probability law at the level of individual detection events rather than ordinary bright-light power measurements.

A possible Stage B question is: Does the measured single-photon transmission probability through a rotatable analyzer follow the predicted $\cos^2$ dependence within experimental uncertainty?

This stage requires specialized equipment and supervision. Do not attempt to build or operate high-voltage photon detectors, lasers, or sensitive quantum-optics equipment without trained laboratory supervision.

14. Optional Stage C — Advanced Research Direction

After mastering polarization measurements, a future laboratory project could investigate polarization correlations between two photons and, under expert supervision, explore a Bell-inequality experiment. This should be treated as a separate advanced project, not as the first experiment.

The research progression is therefore: classical/bright-light polarization measurement → quantitative uncertainty analysis → low-light/single-photon polarization → polarization correlations → advanced quantum-information experiment.

15. Expected Results

The main result should be a graph showing measured intensity versus analyzer angle together with the fitted $\cos^2$ curve and uncertainty information.

A successful experiment is not defined by obtaining a perfect curve. A scientifically valuable result identifies where the data agree with the model, where they deviate, and why.

Possible conclusions include: agreement within uncertainty; statistically significant but explainable deviations; or evidence that the apparatus needs recalibration. All three can be legitimate scientific outcomes.

16. Research Outputs

1. Full research proposal.
2. Laboratory notebook with dated entries.
3. Raw, unmodified data files.
4. Python analysis code.
5. Calibrated/processed dataset.
6. Uncertainty budget.
7. Final research report of approximately 10–20 pages.
8. One-page scientific poster.
9. 5–10 minute presentation.
10. One-page research summary for applications to mentors, summer schools, scholarships, and research programs.

17. Scholarship Strategy

The experiment should be presented as evidence of research readiness, not as a guarantee of funding. A strong application can explain what question you asked, what you built or measured, what data you collected, how you analyzed uncertainty, what you learned, and what advanced research you want to pursue.

Keep evidence: photographs of the setup, dated notebook pages, raw data, code repository, graphs, report, mentor feedback, certificates, and presentation materials. Never fabricate data, results, laboratory access, or authorship.

International quantum programs demonstrate that hands-on research experience is valued. For example, the Chicago Quantum Exchange Open Quantum Initiative has offered undergraduate research experiences in quantum laboratories and research groups, including quantum optics/information science, and the 2026 program included a stipend, housing, and travel for eligible U.S.-enrolled undergraduates. UCL also advertises paid summer quantum research placements for eligible undergraduates. Eligibility varies substantially by program, so these should be treated as examples of the type of opportunity to target rather than promises of eligibility.

18. 12-Week Implementation Schedule

Weeks 1–2: Study polarization, quantum states, Malus' law, uncertainty, and experimental safety.

Weeks 3–4: Obtain/borrow equipment; design mounts; test detector; develop a measurement protocol.

Weeks 5–6: Calibration and pilot measurements; identify major error sources.

Weeks 7–8: Main data collection with repeats and controlled conditions.

Weeks 9–10: Python analysis, curve fitting, residuals, uncertainty budget, and robustness checks.

Week 11: Write the research report and prepare figures/poster.

Week 12: Final presentation, mentor review, corrections, and preparation of scholarship/research application materials.

19. How to Approach a Professor or Laboratory

Use a short message with: your academic background; the project title; what you have already done; the evidence you can provide; the equipment or expertise you need; and a specific request for supervision or access.

Suggested subject: Student Research Request — Experimental Quantum Polarization Project

Suggested opening: Dear Professor, I am developing a student experimental project on quantitative quantum polarization measurements. I have prepared the research question, experimental protocol, uncertainty plan, and Python analysis workflow. I am looking for a supervised laboratory opportunity to perform the measurements with calibrated equipment and, if feasible, extend the work to low-light or single-photon detection. I would be grateful for the opportunity to discuss whether your group could provide guidance or laboratory access.

20. Risk, Ethics, and Scientific Integrity

Follow laboratory safety rules. Never look into a laser beam. Use beam stops and appropriate protective measures. Use only equipment you are trained and authorized to operate.

Do not invent measurements to obtain a beautiful graph. Keep raw data, document failed trials, and explain anomalies.

Do not claim that a bright-light polarization experiment is a loophole-free test of quantum foundations. State precisely what physical prediction was measured and what the apparatus can and cannot establish.

21. Final Project Statement

This project aims to transform an interest in quantum mechanics into a reproducible experimental research portfolio. The initial experiment is deliberately achievable with modest laboratory resources, while the research design provides a clear path toward single-photon and quantum-information experiments when appropriate equipment and supervision become available.

The ultimate objective is not simply to obtain a certificate. It is to demonstrate the habits of a young researcher: asking a precise question, designing a measurement, collecting honest data, quantifying uncertainty, comparing evidence with theory, communicating results clearly, and identifying the next scientific question.

22. Selected References and Further Reading

1. Chicago Quantum Exchange, Open Quantum Initiative Undergraduate Fellowship — information on undergraduate quantum research experiences and eligibility: <https://chicagoquantum.org/oqi-undergraduate-fellowship>

2. UCL Quantum, Undergraduate Summer Placements — examples of paid quantum research placements and laboratory projects:
<https://www.ucl.ac.uk/mathematical-physical-sciences/quantum/study/undergraduate-summer-placements>

3. OpenStax, University Physics / quantum and modern physics materials: <https://openstax.org/subjects/science>

4. MIT OpenCourseWare, quantum physics and related university course materials: <https://ocw.mit.edu/>

5. A standard undergraduate quantum mechanics text should be used for the formal treatment of polarization states, measurement, and probability amplitudes.

Appendix A — Suggested Data Sheet

Angle (°)	Raw signal	Background	Corrected	Uncertainty	Repeat	Notes
0						
20						
40						
60						
80						
100						
120						
140						
160						
180						

Appendix B — Minimum Deliverables Checklist

- Research question and hypothesis
- Equipment list and safety assessment
- Calibration record
- Dated laboratory notebook
- Raw data preserved unchanged
- Python analysis script
- Main $\cos^2$ fit and residual plot
- Uncertainty budget
- Final research report
- Scientific poster
- Short presentation
- One-page scholarship/research summary
- Mentor or teacher review